Hardware-Conscious Performance Engineering of Fainder for Distribution-Aware Dataset Search

Master's Thesis Defense

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August 24, 2026

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Outline

Problem

Distribution-Aware Dataset Search and Fainder

- Raw data not searchable (privacy, licensing, scale) $\rightarrow$ per-column **histograms**
- Predicate class: *"p-th percentile* $\bowtie$ *threshold* τ*"*
- **Fainder** [Behme et al., VLDB 2024]:
  - K-means clusters $\rightarrow$ shared bin grid per cluster $\rightarrow$ precomputed cumulative densities
  - Percentile predicate $\Rightarrow$ **binary search**
- GitTables, 5M histograms: $>$2 orders of magnitude vs profile scan, 1 TB $\rightarrow$ 3.2 GB

This thesis
How does Fainder's query phase run on a modern multi-core server, and what makes it fast?

Fainder in Plain Words

- Histogram = "how many values fall in each range"
- One family, one x-axis: the same question lands at the same position everywhere
- Stored per x-axis point: the fraction of each column's values below it — **sorted**, with IDs
- Sorted = binary-searchable: ∼191 short lists, never 5M summaries
- "median > 20" $\iff$ "fraction of values ≤ 20 is < 0.5": τ picks the list, p picks the cut
- Clusters are independent $\rightarrow$ parallel. Search = few dependent reads; emit = many writes

The Python Baseline

- GIL: one thread at a time; `searchsorted` never releases it
- Thread sweep **flat**: 2% spread at any t
- AoS layout: unread ID next to every value
- Interpreter dispatch on every iteration

10,000 queries, 5M histograms

$\sim$2.4 hours

(8,589 s, any thread count)

- IPC 2.35 vs Rust's 0.96 — yet 378$\times$ slower: 6.2$\times$ the instructions, one effective core
- Rust wins by executing fewer instructions across all cores in parallel, not by winning on IPC

Goal and Scope

- Close the algorithm–implementation gap; query results unchanged
- Rust query engine behind the existing Python API (PyO3); >20 optimisation axes ablated
- Constraint: bitwise-identical results, precision = recall = 1.000, every build
- Out of scope: algorithm changes, GPU, distributed execution

The investigation is the thesis, the port is its instrument
Four hardware ceilings + negative composability + a pre-flight methodology.

Method

Experimental Setup

- 2× Xeon Platinum 8468H (Sapphire Rapids): 96 cores, 105 MiB L3/socket, 1 TB DDR5
- 48 cores per socket → $t \leq 48$ stays on one socket

Dataset	n_{hists}	K_{eff}	hists/cluster	role
c256_10gb	324 K	129	∼2.5 K	small
c256_30gb	997 K	184	∼5.4 K	medium
c256_56gb	5.02 M	191	∼26 K	**primary**
c1024_56gb	5.02 M	610	∼8.2 K	OOD (3.2× **sparser**)

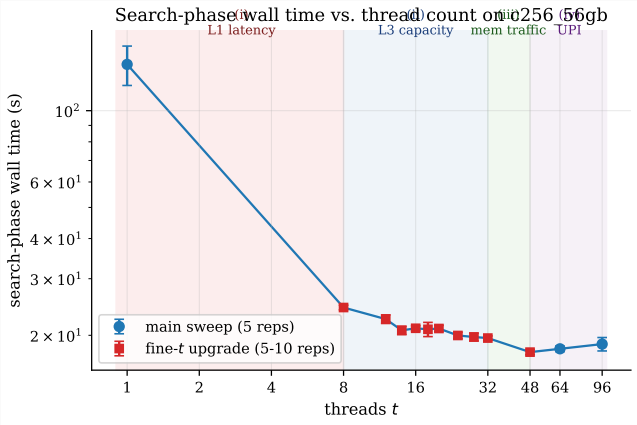
- Same 10K queries everywhere; 5-rep medians; $t \in \{1 \ldots 96\}$; `perf` counters per cell
- 2,658 runs; every claim traces to a benchmark-database row

The Rust Engine

- PyO3 extension module; clustering and loading stay in Python
- Rayon work-stealing over clusters (idle threads take work); no GIL in the hot path
- One Cargo feature flag per axis $\rightarrow$ every variant is a separate measurable build:
  - layout: SoA / AoS, f16, packed IDs search: AVX-512, 8-way batch, lockstep, PGM
  - scheduling: column-centric, query-batch, morsel, pinning memory: pooled, mimalloc, NUMA
- Every build validates bitwise-identical against the reference

Results: Four Ceilings

Finding 1: Four Performance Ceilings



1. Single reads ($t \leq 8$): refuted
2. Shared L3 ($t \approx 16$)
3. Memory-traffic pressure ($t \in [32, 48]$)
4. Cross-socket link ($t > 48$)

One regime each.

One intervention class each.

Ceiling (i): Single-Thread Latency, Refuted

What it is: binary search is a **dependent-load chain** — probe $i{+}1$ cannot start until probe i returns. The literature's default bottleneck for in-memory search.

Ablation	Idea	vs default, all 18 cells
`simd`	vectorise the last ∼16 steps	0.90–1.07× noise
`batch-search`	overlap 8 searches' loads	0.96–1.09× noise
`horizontal-simd`	16 queries walk in lockstep	0.95–1.12× noise
`pgm`	predict the position, skip the chain	0.91–1.13× noise

- Counters at $t{=}1$: IPC $\approx$ 1.9, L1d miss/load 0.5–0.7% — the chains are cache-resident
- PGM verifiably cuts chain depth to $O(\log \log n)$; the wall does not move
- The wall is the *number* of chains, not their depth: 10 K queries $\times$ 191 clusters $\times$ 2 searches $\approx$ 3.8 M chains of ∼22 cheap probes each

Ceiling (ii): Shared L3 Capacity

The claim: at $t \in [8, 32]$ the threads' combined working sets overflow the shared 105 MiB L3. Two probes (< 1 = faster than default, primary workload):

Build	t=1	t=8	t=16	t=32	t=64	t=96
`aos` (capacity probe: doubles bytes/line)	1.03	1.17	1.02	1.00	0.96	0.99
`pooled` (allocator probe: no per-cluster allocs)	0.53	0.66	0.80	0.95	0.93	1.02

- `aos` changes *only* footprint; its penalty sits inside the window (+17% at t=8, $\sim$2$\times$ L3 working set) and vanishes outside $\rightarrow$ capacity is real in $[8, 32]$
- Yet the biggest win in that window fixes the allocator's lock, not capacity — and it fades ($0.66 \rightarrow 0.95$) as ceiling (iii) takes over
- `f16` (halves bytes/value): null on the dense primary, pays $\sim$20% at the sparse OOD density — footprint only matters where it actually binds

Verdict: one regime, two mechanisms — allocator queueing on dense data (few huge clusters), L3 footprint on sparse data (many small clusters).

Ceiling (iii): On-Socket Memory-Traffic Pressure

What it is: all cores share the queues on the path to memory (L3 $\rightarrow$ mesh $\rightarrow$ memory controller). Each added core lengthens every queue, so *every* access waits longer — while total traffic sits at $\sim$0.5% of STREAM peak (the machine's measured maximum memory throughput). **Congestion, not saturation.**

The signature, c256_30gb, wall in seconds:

Build	t=64	t=96		step
default	4.15	5.89	+42%	(IPC 1.93 $\rightarrow$ 0.55, LLC misses $\times$1.6)
`packed-ids`	3.88	4.70		+21% (same step, half the damage)

- Anti-scaling: adding 32 cores makes the run 42% *slower*
- Fix class = fewer bytes per operation: `packed-ids` emits 20-bit instead of 32-bit IDs ($-$37% emit bytes) $\rightarrow$ $-$20% at the anti-scaling cell

Ceiling (iv): Cross-Socket UPI Interconnect

Same workload, four memory placements (`numactl` probe, wall in seconds):

Placement	meaning	t=32	t=48	t=64	t=96
A: unpinned	OS default, first-touch	19.28	16.97	18.19	17.58
B: single socket	cores + memory on socket 0	**16.77**	**15.04**	–	–
C: interleave	pages alternate sockets per 4 KB	25.87	23.73	22.91	24.83
D: cross-socket	data socket 0, threads socket 1 (all remote)	19.08	16.86	–	–

- B vs A: removing all cross-socket traffic saves 11–13% — that saving *is* the link cost
- C vs A: "fair" page balancing costs 26–41%: half of all accesses land remote, and the per-page alternation defeats the prefetcher
- D $\approx$ A: even the worst case matches default $\rightarrow$ the baseline was already paying the link

Results: Negative Composability

Finding 2: Optimisations Compose Negatively

`best-combo` = six features, each regime-best somewhere, bundled (`pooled f16 simd pin-cores cluster-prefetch mimalloc`): 0.773× default wall at t=16. One more optimisation on top:

Composition	Intended axis	Outcome
`best-combo` + `packed-ids`	emit-ID bandwidth	NEG 12/18 cells, up to +24%
`best-combo` + qbatch K=128	cold-load amortisation	NEG every cell, +8 to +32%
qbatch + morsel	cross-worker coordination	NEG every cell, +8 to +12%
`packed-ids` + `local-ids`	read-side bandwidth	NEG +186 to +245%
first-touch + interleave	memory-locality balance	NEG every cell, +26 to +41%

qbatch: batch K queries per cluster visit morsel: cross-worker coordination `local-ids`: per-cluster renumbering

- One shape, five axes: the coarser optimisation absorbed the headroom, the finer one's overhead turns net cost — every plausible positive argument falsified by `perf`. Structural, not coincidence

Finding 3: Pre-Flight Ceiling Identification

Classical ablation (build $\rightarrow$ measure vs default $\rightarrow$ keep if faster) assumes optimisations are independent. With stacked ceilings, whether B helps depends on the binding ceiling *at the target composition*.

Before writing code:

1. Probe the binding ceiling at the target composition (`perf`)
2. Axis no longer binds there? Stop
3. Predict the saving (bandwidth-budget arithmetic)
4. Measure where the ceiling binds; separate saving from overhead
5. A negative composition counts as a result

Machine-agnostic: applies to any hardware-near system with stacked ceilings.

Dispatch on Regime

- No dominating build → per-regime selection: (n_{hists}, t) → build + flags
- `best-combo` at moderate t / big data; `packed-ids` at t=64; qbatch at t=96; columnar at low t; default on small data
- In-distribution: 17/18 exact, 18/18 within 5% — combined with OOD: 23/24

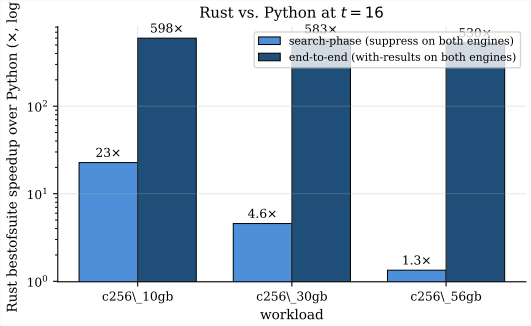
Out-of-distribution validation (c1024_56gb, 3.2× sparser, never tuned on):

t	policy pick	regime-best	gap
1	qbatch K=64	qbatch K=128	+0.9%
8–32	`best-combo`	`best-combo`	0.0%
64	`packed-ids`	`best-combo`	+0.2%
96	qbatch K=128	qbatch K=128	0.0%

Boundaries are machine-specific; the probing procedure is not.

Results: Comparison with Python

Speedup over the Python Baseline



Workload	search	end-to-end
c256_10gb	22.7×	598×
c256_30gb	4.6×	583×
c256_56gb	1.34×	530×

- End-to-end: 2.4 h → 16 s
- ∼99% of the ratio = avoiding Python's result-dict construction
- Search alone bends toward the shared ceilings
- Rebinning kernel: 4.7× / 2.2× on construction

Related Work

Positioning Against the State of the Art

- Layouts: Manegold et al. [2000] — memory access dominates → our SoA; ART, Eytzinger, learned indexes target cache behaviour
- On heterogeneous, cache-resident arrays: classical binary search wins (consistent with ceiling (i))
- Execution: Neumann [2011] compile-don't-interpret → the port; Leis et al. [2014] → work-stealing
- Composability traditions (Selinger [1979]; Caladan [2020]; memory-wall) assume gains compose
- No prior documentation of five mechanism-grounded negative pairs + a falsifiable pre-flight check
- Gap filled: first hardware characterisation of histogram-based percentile search; first horizontal-SIMD measurement on Sapphire Rapids

Conclusion

Contributions

1. Rust re-implementation: bitwise-identical, 530–598× end-to-end, PyO3-integrated — *engineering*
2. Four-ceiling characterisation on Sapphire Rapids, via `perf` — *scientific*
3. Negative-composability pattern: five instances, five axes — *scientific*
4. Pre-flight ceiling identification — *methodological*
5. Dispatch-on-regime policy, 23/24 within 5% incl. OOD — *engineering*

(1) and (5) are the staging ground; (2)–(4) generalise.

Future Work

- (F1) Transparent Huge Pages: 512× fewer TLB misses per covered page → targets the local-ids mechanism
- (F2) NUMA-aware build for $t > 48$: shard clusters across sockets, node-local workers
- (F3) Adaptive precision: f32 at low t, f16 at high t / sparse density, selected at runtime
- Column-centric: widen its t=1 win (2.17–2.63×) by fixing cross-cluster imbalance
- Dispatch validation beyond the ∼8.2 K–26.3 K hists/cluster range

Closing Remark

- The speedups are specific to this codebase, machine, and workload
- What generalises: whether optimisation B helps depends on which ceiling A already collected — invisible to per-operator cost models

Closing remark
Identify the binding ceiling at the target composition before committing the implementation cost of the next optimisation.

Questions?

Backup

Backup: One Query, End to End

Query: *"median $>$ 20"* = (p=0.5, $>$, τ=20)

median $> 20 \iff$ CDF(20) < 0.5

Cluster A index (toy: 3 histograms)

bin edges: $[0, 10, \mathbf{20}, 30, 50]$

edge	sorted CDFs	IDs
$\leq$10	$[0.00, 0.10, 0.20]$	H2 H1 H5
$\leq$**20**	$[\mathbf{0.25}, \mathbf{0.45}, 0.50]$	**H2 H5** H1
$\leq$30	$[0.75, 0.75, 0.80]$	H2 H5 H1

1. Binary search τ=20 in the bin edges $\rightarrow$ column "$\leq$20"
2. Binary search 0.5 in that sorted column $\rightarrow$ cut after 2 entries
3. Emit the IDs before the cut: H2, H5
   (H1 sits exactly at 0.50: median $= 20$, not > 20)
4. Cluster B (salaries, range $\geq 10\,000$): τ below range $\rightarrow$ **all match, no search**
5. Union across clusters = the answer

Clusters are independent $\rightarrow$ the cluster loop parallelises. Search = few dependent reads; emit = many writes.

Backup: Why is end-to-end 530× if search is only 1.34×?

- Python's per-(query, cluster) result-dict construction dominates its wall. Measured at t=16, materialisation accounts for:
  - c256_10gb: 580.9 s of 599.7 s (96.9%)
  - c256_30gb: 1,878.5 s of 1,890.3 s (99.4%)
  - c256_56gb: 8,568 s of 8,589 s (99.8%)
- Rust emits results without per-cluster dict construction, so it avoids this cost entirely.
- We therefore report both columns: suppress-vs-suppress isolates the search engines, with-results-vs-with-results is what a user experiences.
- Conflating the two would attribute Python's materialisation cost to Rust's search engineering.

Backup: Why does f16 win nothing standalone?

- Prediction: halve the footprint, win in the regime where the f32 working set exceeds L3 but the f16 one fits.
- On c256_56gb the per-cluster f32 slice (∼105 KiB) is already L2-resident. Halving it unlocks no new cache-resident regime, while every comparison pays the f16→f32 dequantisation.
- Result: no significant standalone win in the 10-rep fine-t sweep, and a +15.6% drag inside the bundle at t=16.
- On c1024_56gb (3.2× sparser, ∼33 KiB slices, 610 clusters) the aggregate cluster-loop footprint binds. There f16 carries ∼20% of the bundle win.
- Footprint optimisations are density-conditional.

Backup: Why “memory-traffic pressure” and not “DRAM bandwidth”?

- STREAM Triad on this machine reaches $\sim$330 GB/s aggregate.
- Accounting the workload’s LLC-miss traffic at 64 B per missed probe puts it near 0.5% of that peak.
- “Bandwidth-bound” claims saturation of the DRAM channels. The counters show nothing close to saturation.
- What we observe is anti-scaling from aggregate pressure on the on-socket memory subsystem: queue occupancy and allocator traffic, at utilisation far below peak.
- The name matters because it predicts which interventions help: fewer bytes per comparison and per emit. A pure bandwidth upgrade would not.

Backup: The local-ids regression

- `local-ids` (per-cluster reindexing) was pre-registered as a positive prediction: narrower IDs, less read-side bandwidth.
- Measured: $+186$ to $+245\%$ at $t \in [8, 64]$ on top of `packed-ids`. The prediction is falsified.
- Mechanism, from a direct `perf stat` probe: the bw$\approx$13 decoder's access pattern over the packed buffer raises the dTLB miss rate $52\times$ per instruction.
- TLB-walk stalls at $\sim$100 cycles each drown the bandwidth saving.
- Future work (F1): Transparent Huge Pages (2 MiB pages) cut TLB-miss frequency $512\times$ per covered page. That probe is the natural follow-up.

Backup: What generalises beyond this machine?

- The thresholds do not generalise. The methodology does.
- All experiments ran on one Sapphire Rapids 8468H configuration. Genoa, Granite Rapids, or Grace would shift per-ceiling thresholds and dispatch boundaries.
- The pre-flight procedure (probe the binding ceiling at the target composition) is machine-agnostic. The four-ceiling taxonomy is a hypothesis set to re-test on new hardware, and not a constant.
- Density scope: dispatch is validated between $\sim$8,200 and $\sim$26,300 hists/cluster. Outside that range the policy is a prediction.
- A multi-machine deployment would add a fifth candidate ceiling (network) with its own intervention class.

Backup: How was correctness validated?

- Every Rust build validates bitwise-identical against the Python reference across all workloads: precision = recall = 1.000.
- Validation runs inside the benchmark harness, so each measured configuration is also a correctness check. Suppress mode is a stopwatch setting only; correctness is always checked with results on.
- Rebinning kernel: verified against the Python reference on 200 queries. Rust rounds half-away, NumPy uses banker's rounding. The maximum cumulative difference is 10^{-4} at a handful of rows, with no effect on sort order or query results.
- The kernel falls back to the Python pipeline if the extension is not importable or for `cubic_spline` bin estimation.

Backup: Why Rust, and not C++, Cython, or Numba?

- The requirements: no GIL in the hot path, explicit control over layout and allocation, safe shared-memory parallelism, and a maintainable Python integration.
- Rust with PyO3 covers all four: Rayon work-stealing without data races, Cargo feature flags that make every ablation a separate reproducible build, and a drop-in extension module behind the existing API.
- C++ could reach the same performance. The feature-flag ablation matrix and memory safety under 96-thread parallelism are where Rust paid off in practice.
- Numba and Cython remove less of the interpreter overhead and leave allocator and layout decisions with Python. Our measurements show those layers are the actual cost.

Backup: Why does the rebinning speedup drop at 5M histograms?

- 4.72× (324 K) and 4.77× (997 K) drop to 2.18× at 5.02 M histograms.
- Cause: the kernel's own parallelism ceiling. At that scale it runs at ∼2 effective cores of 96, with ∼46% of kernel time on a shared per-cluster allocation.
- An earlier hypothesis (NumPy bulk operations dominating on the Python side) was tested and refuted.
- This is a construction-time cost, paid once per index build. The query-phase findings are unaffected.